Art & Science physics, grad program

Learning Aesthetics: Comparing Vision and Vision–Language Embeddings for Human Art Ratings

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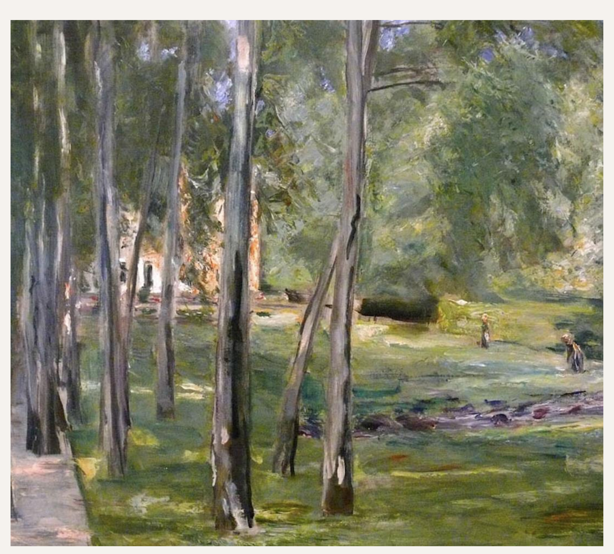
Motivation

- Mosaic artworks consist of many small elements arranged into structured patterns that elicit a strong and distinctive visual experience.
- We seek to understand how observers form aesthetic judgments for such abstract mosaics—what makes some configurations more pleasing than others and whether machine-learning models can predict these human ratings.
- Motivated by prior evidence that the brain's language systems contribute to the interpretation of visual stimuli, we compare the performance of vision-only and vision–language models in predicting aesthetic evaluations.

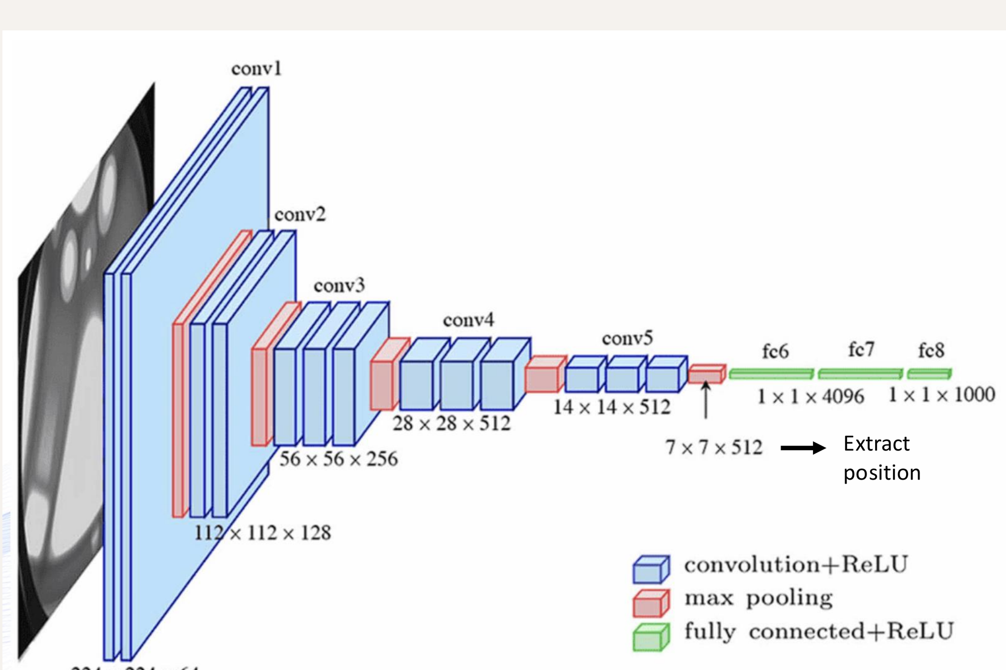
Methods / Data

- Our database, provided by a collaborator, contains over 800 artworks spanning both abstract and narrative styles, each accompanied by a human aesthetic rating.




- **Machine-learning models we use :**
  Vision-only encoders: AlexNet, Inception, ResNet50, VGG16, ViT-B/16
  Vision–language encoders: BLIP2, CLIP-B16, CLIP-B32, ImageBind, SigLIP2.

- **Embedding extraction**
  For each artwork, we feed the image into each pretrained model with its official preprocessing and extract a single global feature vector

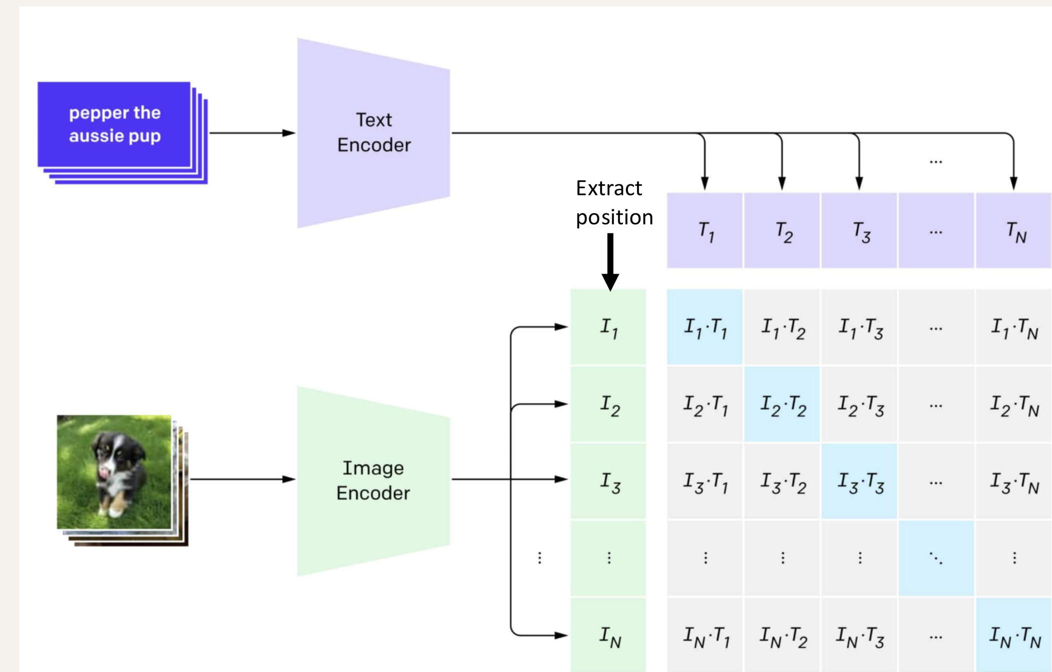


VGG-style convolutional neural network architecture (image from Gaudenz Boesch, *Very Deep Convolutional Networks (VGG) Essential Guide*, viso.ai, 2021).

CNN-based models: take the 7×7×512 feature map from the final convolutional layer

↓

Global average pooling (GAP)
1 x 1 x 512



Schematic of a modern vision–language model architecture (image adapted from Gaudenz Boesch, *Vision Language Models: Exploring Multimodal AI*, viso.ai, 2024).

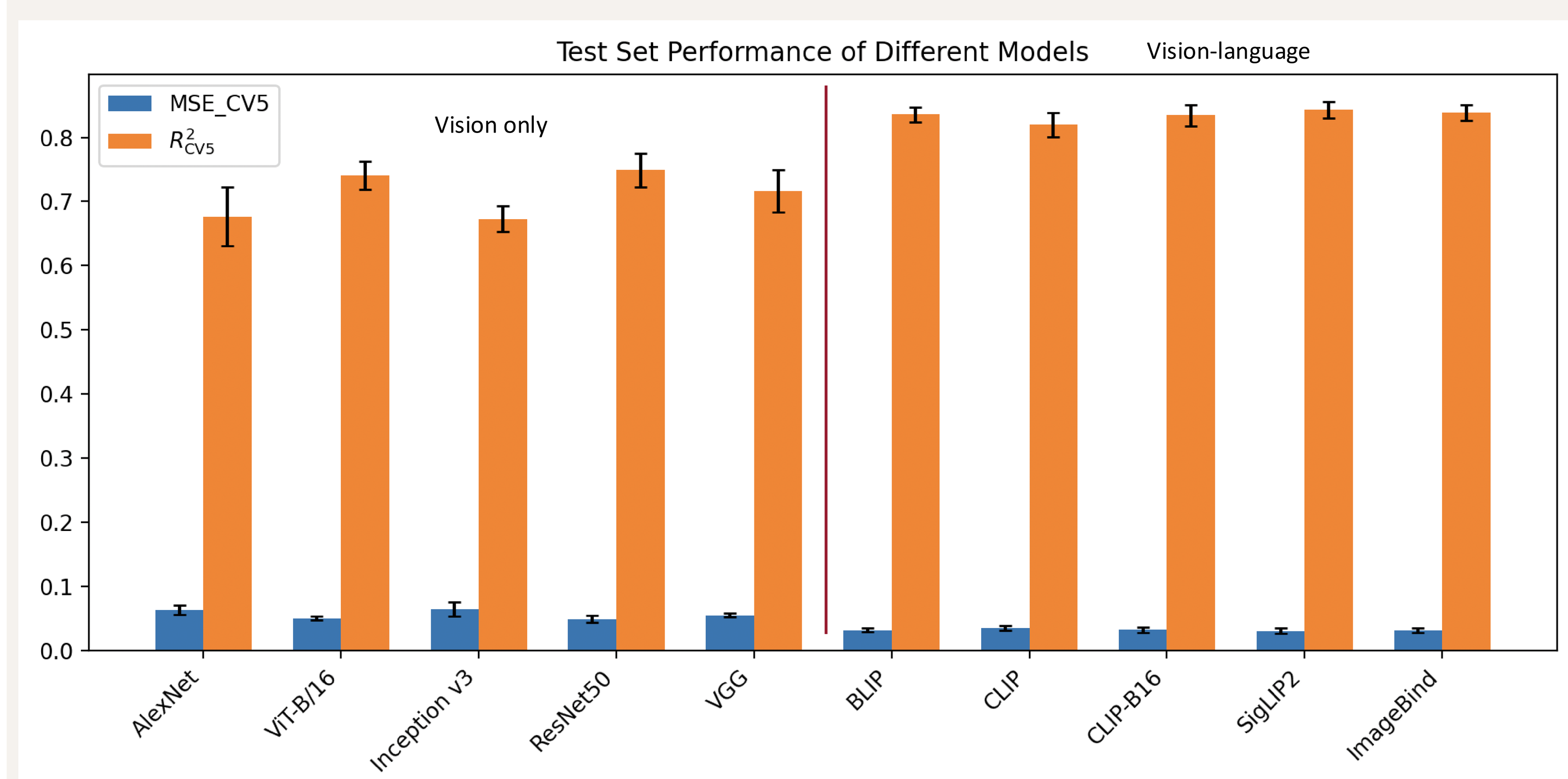
For vision–language models, we take the pooled image embedding from the image encoder

Linear regression & outputs

- **Regression :**
  For each model, we form a matrix X of image embeddings and a target vector Y of human ratings, then fit a Ridge regression model (with L2 normalization) on an 8:2 train–test split.

  We use mean squared error (MSE) and R^2 as our metrics: MSE is the average squared prediction error (lower is better), and R^2 is the fraction of variance in human ratings explained by the model (higher is better).

- **Result :**



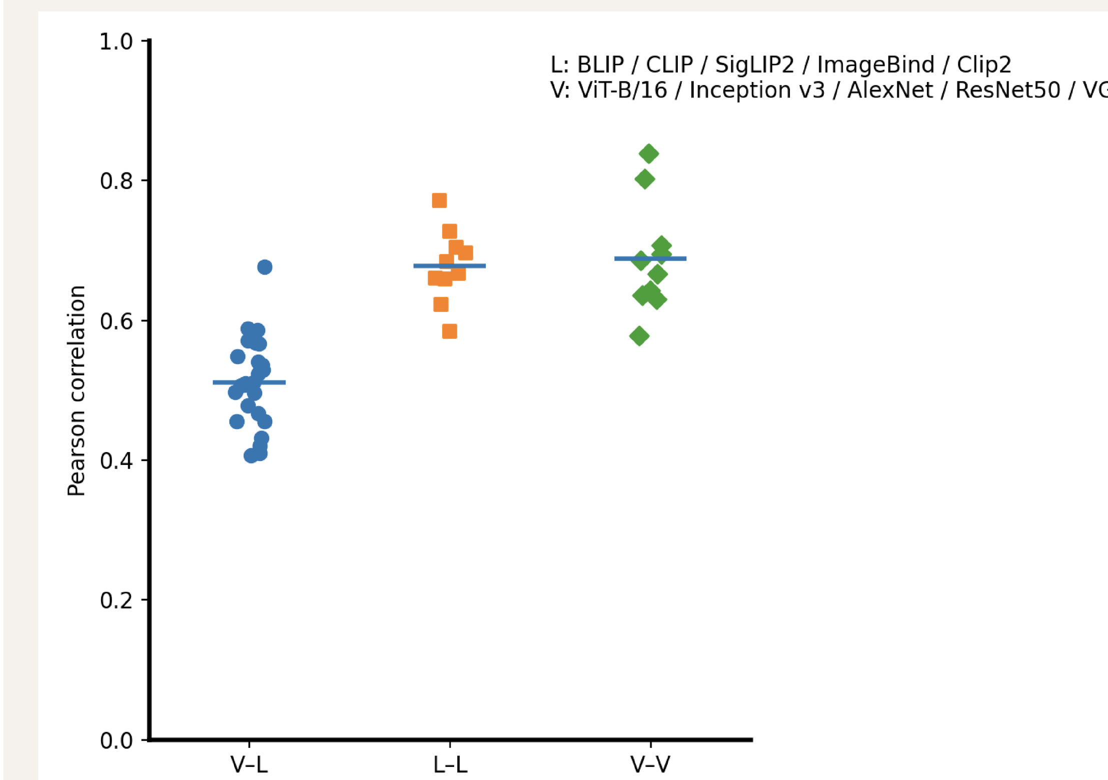
Five-fold cross-validated performance of vision-only models (left six bars) and vision–language models (right five bars). Blue bars show mean MSE_{CV5} (lower is better) and orange bars show mean R^2_{CV5} (higher is better); error bars indicate ±1 standard deviation across folds.

Vision–language models achieve lower MSE and higher R^2 than most vision-only encoders.

Among vision-only models, ResNet50 is the strongest and approach the best vision–language models.

Error patterns & Pearson correlation

- We ask whether different models make similar or different mistakes. For each pair of models, we compute the Pearson correlation between their per-image prediction errors and then group these correlations by pair type: V–V (vision-only vs. vision-only), L–L (vision-language vs. vision–language), and V–L (mixed pairs).



Pearson correlations of per-image errors grouped by pair type. Each point is one model pair; horizontal bars show the mean correlation per group.

Pearson correlation: how similarly two models' errors vary across images:
- closer to 1 means they tend to the same errors
- closer to 0 means their errors are more independent.

- This figure shows that vision-only and vision–language models have systematically different error patterns, indicating that the two families do not behave identically and may be partly complementary.

Discussion & Future work

- **Sanity check (random shuffle)**
  We randomly shuffled which rating goes with which image and reran the training. After shuffling, all models' R^2 scores dropped close to 0. This shows that our pipeline is really learning the relationship between images and ratings, not just noise or artifacts.
- **Discussion:**
  With the true ratings, both vision-only and vision–language models can predict human aesthetic scores quite well, with vision–language models achieving the best overall performance. At the same time, the error correlations show that these two model families do not behave in the same way.
- **Next step:**
  We will dig deeper into why vision-only and vision–language models differ by analyzing which features each family relies on and where their predictions diverge.

Boesch, G. (2021). *Very Deep Convolutional Networks (VGG) Essential Guide*. viso.ai. Retrieved from https://viso.ai/deep-learning/vgg-very-deep-convolutional-networks/
Boesch, G. (2024). *Vision Language Models: Exploring Multimodal AI*. viso.ai. Retrieved from https://viso.ai/deep-learning/vision-language-models/